



A new DSWTS algorithm for real-time pedestrian detection in autonomous agricultural tractors as a computer vision system



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ABSTRACT

This study presents the results of a human detection algorithm using a single camera installed in front of the tractor. An algorithm was designed and implemented in five stages named DSWTS (Division, Segmentation, Watershed techniques, Thresholding, Subtraction). The algorithm first changes an input RGB image to a grayscale image. This image is then divided into small blocks. So, deletion of some unneeded regions makes image processing more comfortable. To segment an object from the background, the edges of the object are detected using a magnitude gradient function and watershed techniques. Then, by subtracting human from background and comparing series of image frames, the pedestrian is recognized. The algorithm was evaluated under morning, noon and evening lighting conditions. Its results were compared with the histogram of oriented gradient (HOG) method and the cascade method that are commonly used to identify humans in images. The results show that the DSWTS algorithm has good accuracy at 8–20 m. Also, in order to improve its performance from 0 to 8 m distances, it combined with two other algorithms. Then by comparing and evaluating combined algorithms, best fusion was discovered and found good results.

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1. Introduction

Machine vision applications vary according to the technique used. Each image contains noise and deformation. Humans do not need to filter and remove noise to understand what is seen, but machine vision requires this to process an image. One task of vision is to detect and track humans under different environmental conditions. It is difficult to track a human as the object under uncontrolled lighting conditions, in a crowded environment, or when the camera is in motion. These conditions can cause loss of information from the received image, such as the color, texture and shape of the object, which can be important to successful detection of the target.

Most methods of target detection use image indices [1–4]. Pedestrian detection is usually done in two steps: feature extraction and classification. Feature extraction is preferred over the direct use of the raw pixels from images. One reason is that feature-based systems are quicker than raw pixel-based systems [5]. In the classification stage, feature-based systems first extract the features and the performance of the classifier is highly dependent

on the feature extracted. The proper selection of feature extraction and classification methods improves the detection rate.

Haar-like features are composed of a number of black and white areas for which the intensities of the pixels in the black and white areas are calculated separately. The feature value is obtained using a weighted combination of the two values. Next, an adaboost detection algorithm is employed and then cascade classification is used. Initial classifiers are used to remove most negative samples. Next, classifiers that achieve low false positive error rates with increased computation are used. This cascade structure can take less time to achieve a high detection rate [6].

The most vulnerable pedestrians are those in the transportation system. Accurate detection of pedestrian traffic is very important for computer vision in intelligent transportation systems and driver-assistance safety systems [7]. These pedestrian detection systems are usually single-camera based on a non-linear support vector machine (SVM) in which features based on Haar wavelet coefficients are used to evaluate the classification function.

A similar structure is used for face and car detection that is not related to motion, tracking, background subtraction, or hypotheses based on the structure of the view [8]. The location of pedestrians can be determined using four examples based on the detectors. The detectors evaluate four members of the human body (head, legs,

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left arm and right arm) separately. It then ensures that these members form the correct geometry based on classification that categorizes the target object as humanoid. The results are better when the joined body is confirmed [9]. Similar work has been done using stereo vision [10].

Some techniques are based on features such as similarity, the figure of the human body, or features of a vertical line [11,12]. Determination of pedestrian movement patterns using a SVM classification function can also detect humans [13]. Researchers also use expensive hardware such as optical cameras, infrared cameras and conventional radar to provide real-time detection systems, but the use of these on automobiles is not generally cost-effective [14].

Networks of histogram of oriented gradient (HOG) descriptors using linear SVM classification have good result in images that are to detect humans [15,16]. Humans can also be detected using an algorithm with low-level features to discriminate between pedestrians and non-pedestrians that can be further evaluated using an adaboost classifier [17].

Some studies in this field use a two-stage algorithm that employs a model for pedestrian detection based on machine vision. The first step uses adaboost and cascade algorithms to segment the pedestrian from the image. The second stage removes errors to confirm that each detected target is the image of the human. At this stage, SVM classification evaluates pedestrian detection and input features using both grayscale and image-edge extraction, the results are better than one-stage classifications like adaboost or SVM [7]. These methods continue to show shortcomings in the application of machine vision to identify pedestrians. Advances in machine vision can be useful for this.

The proposed DSWTS method is a five-stage algorithm, toward achieving this goal. We have named this algorithm as DSWTS by using first letter of each stage (Division, Segmentation, Watershed techniques, Thresholding, Subtraction). This algorithm is faster and more accurate in farther distances (8–20 m) in comparison with the other methods that were talked about them above as well. As can be seen, an application based on a computer vision system for pedestrian detection in tractors is efficient if it can work in farther distances (farther than 8 m to prevent accident). So, with due attention to performance of the proposed DSWTS algorithm at distances of 8–20 m when the human is going out of the frame, or in poor light conditions (shade), the previously proposed HOG and cascade methods were not accurate at these distances and lighting conditions. Also, to produce an overall algorithm that is highly accurate at all distances under all lighting conditions, combinations using two-by-two and all three algorithms were carried out too.

The remainder of the paper is organized as follows. Section 2 explains the materials and methods to discuss the proposed algorithm. The different stages of the proposed DSWTS algorithm and

different features used to pedestrian description are defined. Section 3 first describes the results of the proposed algorithm in the different distances and light conditions, then compares it with the HOG and cascade methods at these distances and light conditions, finally describes the results of experiments by using different combination algorithms to find overall algorithm. Section 4 presents the conclusions and explains which one of these algorithms is optimal with acceptable performance at all legal distances and different lighting conditions.

2. Materials and methods

In this study, a single CMOS camera (Logitech c920 HD pro webcam) with 640×480 pixel resolution at 30 frames/s is used and installed on top of an MF399 tractor and connected to a computer using wires. The system is shown in Fig. 1. The camera was controlled using MATLAB software and the settings were adapted to send color images of the scene in front of the moving tractor to the computer.

The image-processing toolbox in MATLAB applies, in the image acquisition section by doing settings like device selection (when more than one camera connected to a computer, the camera can be selected in this section), determining desired video format as MPEG 4 and resolution and recording speed in frames per second, determining the type of capturing the image that is placed in Immediate mode and determining the number of captures and finally specify the number of frames per capture image which is an infinite mode setting. At this point, image acquisition section is done and ready to send to the image processing unit. Given that the standard of acquisition rate of the single camera used in this study was 30 frames per second, so, each image acquisition time was from 10 to 15 s. As a result, the numbers of frames in each video were 300–450 frames. We have selected 10% of the whole frames. In the other word, for each ten frames, one frame was selected to be processed. Fig. 2 diagrams the process in five stages.

2.1. Divide to small blocks and deletion of top right areas

To improve the efficiency of the algorithm, based on the previous researches about the segmentation of objects in image, and also about the existence of different colors of man coverage, skin color, and background color variation per frame of the image, it was concluded that the color index would not be useful in this case. As a result, to avoid more complex and time-consuming image processing operations, it was decided to omit the color index. Doing this, not only makes the image processing more comfortable, but also leads to reduce image processing time as well. Therefore, the color (RGB) image is then converted to grayscale



Fig. 1. Installation location of the camera.

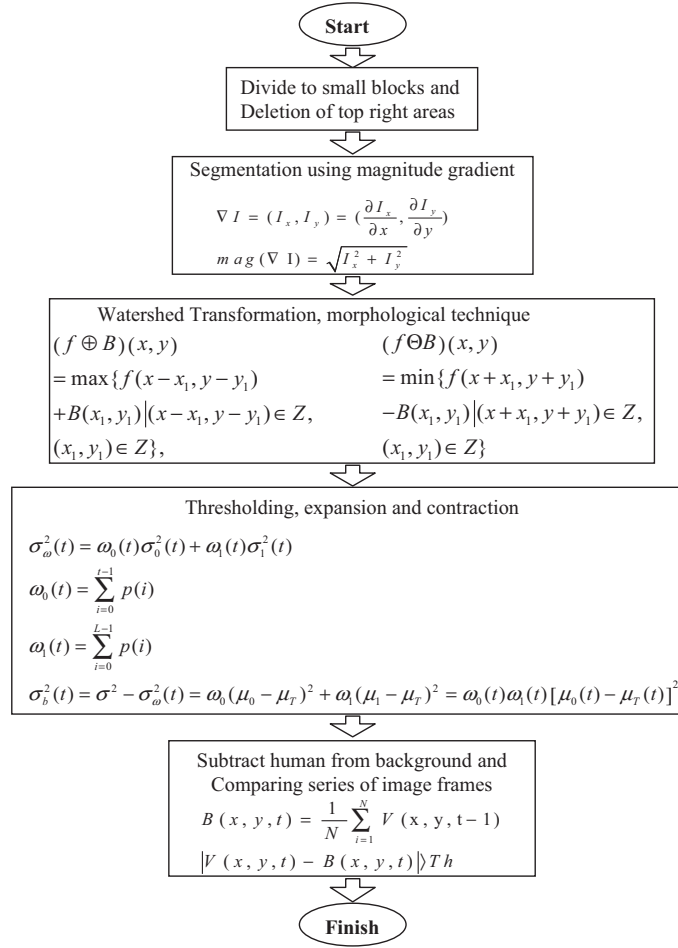


Fig. 2. Block diagram of proposed algorithm.

(Fig. 3). The image is then divided into small blocks and unneeded sections, such as the edge of the road near the tractor, are deleted. These sections do not aid recognition of the human as the object and do not affect immediate control of the tractor. Deletion of the upper edge of the image also improves image processing.

2.2. Segmentation using magnitude gradient

The next step uses the magnitude gradient function to segment the image. The sobel function is used to detect edges, and then filtering and simple calculations are used to calculate the magnitude gradient using Eq. (1) as:

$$\nabla I = (I_x, I_y) = \left(\frac{\partial I_x}{\partial x}, \frac{\partial I_y}{\partial y} \right) \quad (1)$$

$$\text{mag}(\nabla I) = \sqrt{I_x^2 + I_y^2}$$

where I denotes the original image and ∇ denotes the gradient operator.

Fig. 4 shows the resulting image.

2.3. Watershed techniques

The direct use of a watershed transform function (Fig. 5) on the image of the gradient is not possible. Thus, a number of pre-processing operations were performed that these operations are in fact morphological closing operation on the opening operation, which are useful for noise removal. This combination of morphological operations has a role as a low-pass filter. The calculation marks on the human and the background image improve segmentation.

The image impurities were cleared using the morphological function to flatten a maximum region containing the target object. The “imopen” function is used to remove particulate noise during

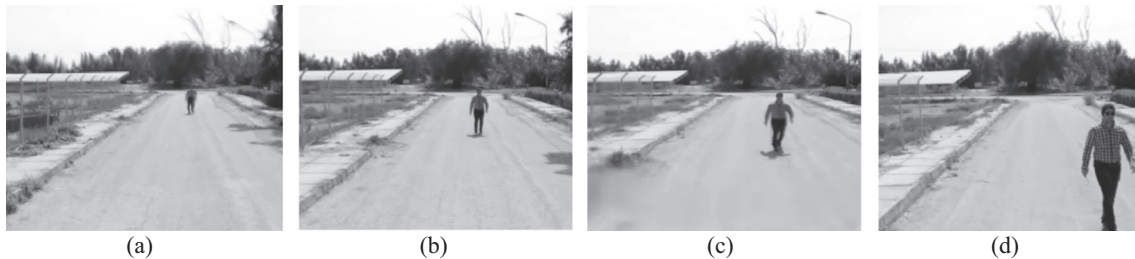


Fig. 3. Grayscale images obtained from RGB images: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m.

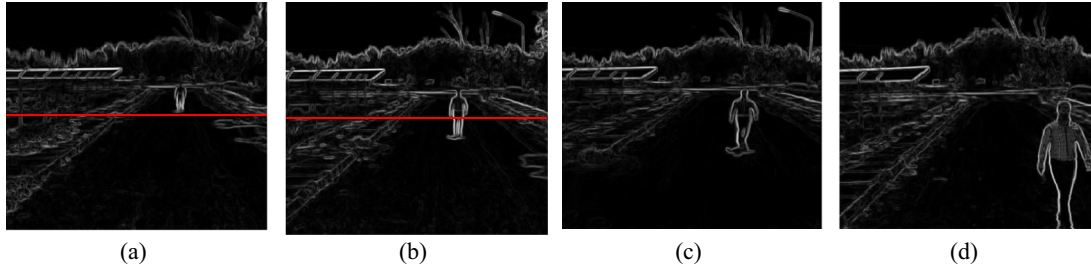


Fig. 4. Magnitude of gradient of images: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m.

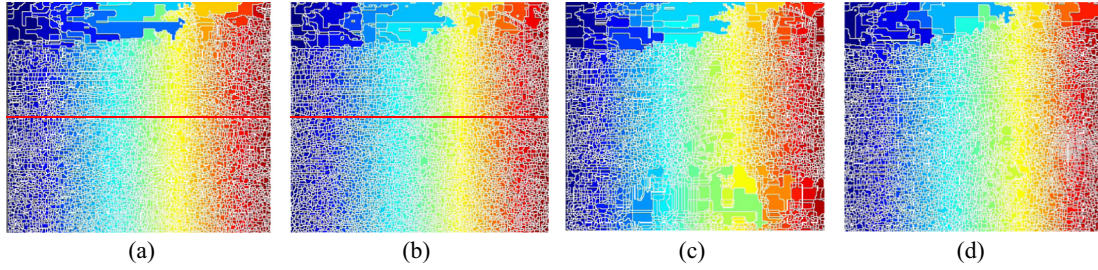


Fig. 5. Watershed transform function of images: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m.

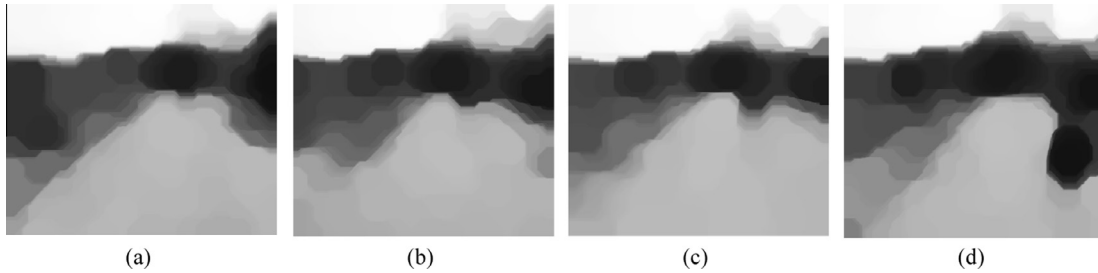


Fig. 6. Image erosion using morphological reconstruction: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m.

this process. Erosion of the image using morphologic reconstruction removes dark particles and noise (Fig. 6). This is more effective than conventional methods and does not affect the shape of the target object. Some of calculations which are used for this reason have shown in Eq. (2) as:

$$\begin{aligned}
 (f \oplus B)(x, y) &= \max\{f(x - x_1, y - y_1) + B(x_1, y_1) | (x - x_1, y - y_1) \in Z, (x_1, y_1) \in Z\}, \\
 (f \ominus B)(x, y) &= \min\{f(x + x_1, y + y_1) - B(x_1, y_1) | (x + x_1, y + y_1) \in Z, (x_1, y_1) \in Z\}
 \end{aligned}
 \quad (2)$$

where f is the image and B is the structuring element, and $\oplus$ and $\ominus$ denote the dilation and erosion operators, respectively.

Next, the maximum highlight region of the target object is calculated to obtain an appropriate image of a human (Fig. 7). To better interpret the results, the foreground marker image has been superimposed on the original image. It is evident that many objects, and absorbed and shadowed areas are not marked, which means that these objects will not be well-segmented from the background. Some of the mostly-occluded and shaded objects are not marked. These objects will not be segmented properly. The foreground markers for some objects go right up to the edge of the object. This indicates that the edges of the marker blobs should

be cleaned and then shrunk. This can be done by closing, followed by erosion. To avoid ruptures, stray isolated pixels must be removed. This operation continues until a specific number of pixels are removed.

2.4. Thresholding, expansion and contraction

Since the dark pixels in these images belong to the background, to mark the background of the image obtained in the preceding steps, Otsu's thresholding is used to convert the image to a binary image (Fig. 8(a)). Eq. (3) shows some of Otsu's thresholding calculations:

$$\begin{aligned}
 \sigma_w^2(t) &= \omega_0(t)\sigma_0^2(t) + \omega_1(t)\sigma_1^2(t) \\
 \omega_0(t) &= \sum_{i=0}^{t-1} p(i) \\
 \omega_1(t) &= \sum_{i=t}^{L-1} p(i) \\
 \sigma_b^2(t) &= \sigma^2 - \sigma_w^2(t) = \omega_0(\mu_0 - \mu_T)^2 + \omega_1(\mu_1 - \mu_T)^2 \\
 &= \omega_0(t)\omega_1(t)[\mu_0(t) - \mu_T(t)]^2
 \end{aligned}
 \quad (3)$$

where σ denotes the standard deviation and μ denotes the averaging operations.

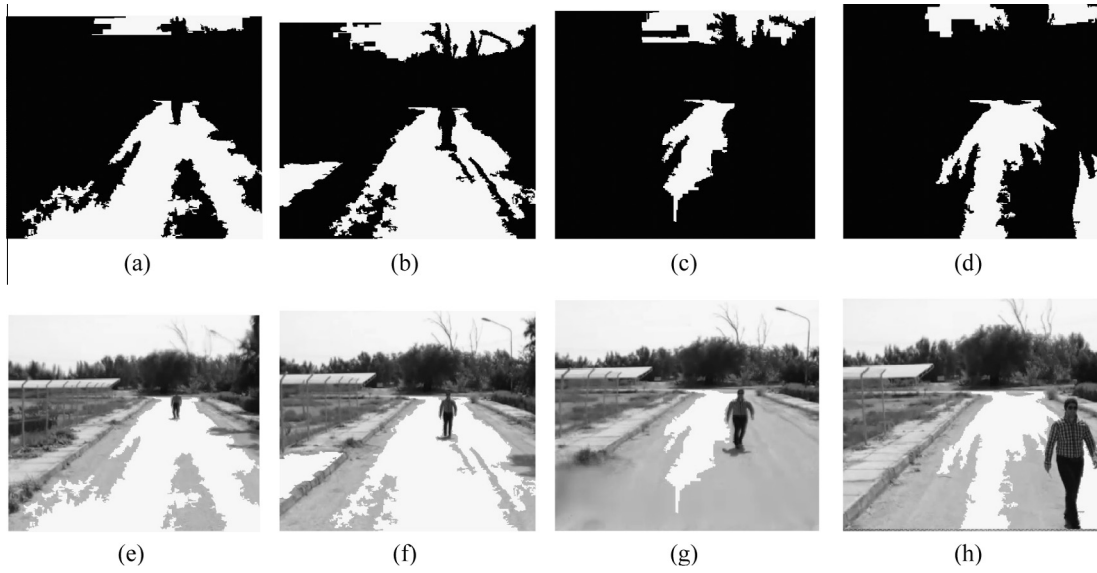


Fig. 7. Maximum highlight region of object: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m; maximum highlight region of object superimposed on original images: (e) Distance: 18 m. (f) Distance: 14 m. (g) Distance: 10 m. (h) Distance: 2 m.

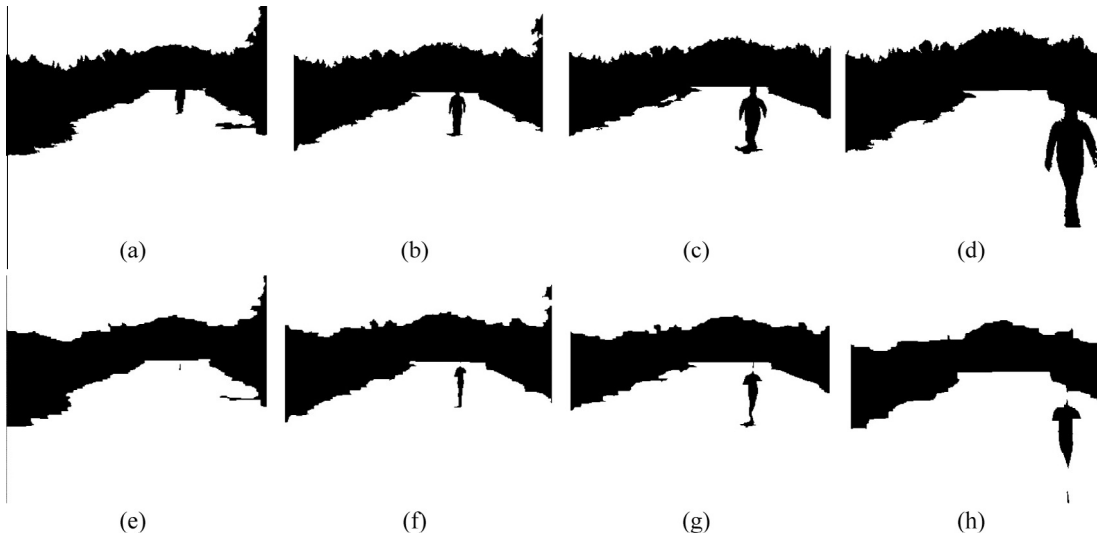


Fig. 8. Binary image after Otsu's thresholding: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m. Contraction of pixels in horizontal and vertical directions: (e) Distance: 18 m. (f) Distance: 14 m. (g) Distance: 10 m. (h) Distance: 2 m.

As shown, the resulting image has a few cuts and ruptures and the head of the target object (human) is in contact with the background. Separation of human from the background and removal of ruptures was done by expansion and contraction of pixels in the horizontal and vertical directions (Fig. 8(b)).

2.5. Subtract human from background and comparing series of image frames

By subtracting an empty background image (Fig. 8(b)), the human can be identified and detected (Fig. 9(a)). The weighted center of the target object (human) has been determined and a box is drawn around the target to identify it as human (Fig. 9(b)). Some of calculations which are used for this reason has shown in Eq. (4) as:

$$B(x, y, t) = \frac{1}{N} \sum_{i=1}^N V(x, y, t - 1) \quad (4)$$

$$|V(x, y, t) - B(x, y, t)| > Th$$

where B is the background and V is the pedestrian (object).

Since in the evaluation of the proposed algorithm, we have processed all existing frames, it was observed that the proposed algorithm had good results in the majority of frames. As a result, we did not use a specific limitation to select the frames. Also, given that the autonomous guidance of vehicles requires fast and real-time response, so, we have chosen one frame randomly from each ten frames. In the other word, only one frame from ten frames is processed. As a result, by reducing the number of frames to be

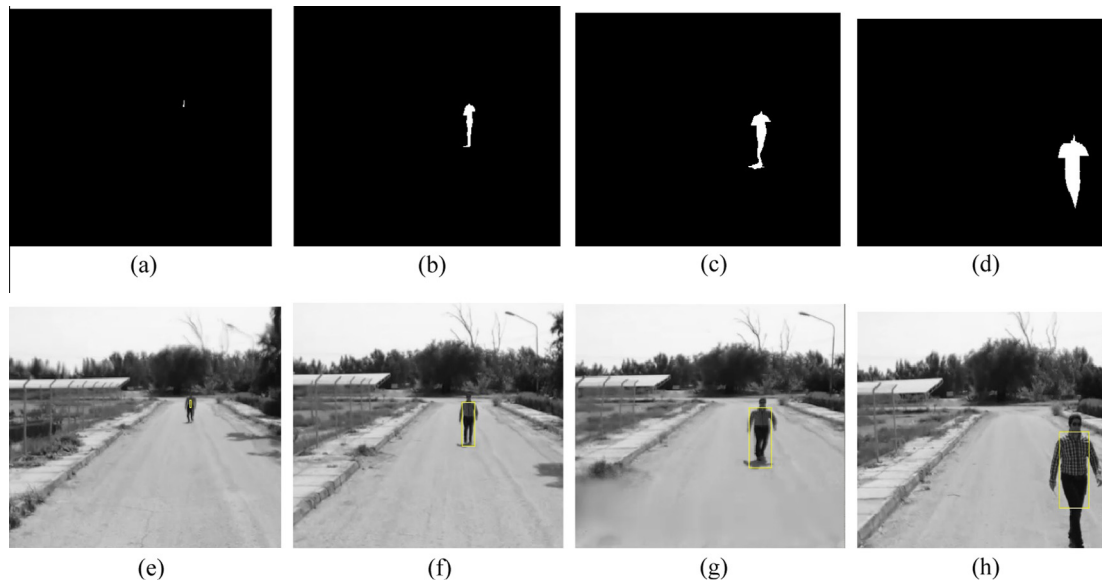


Fig. 9. Image obtained by subtracting background: (a) Distance: 18 m. (b) Distance: 14 m. (c) Distance: 10 m. (d) Distance: 2 m. Image obtained by final identification of human: (e) Distance: 18 m. (f) Distance: 14 m. (g) Distance: 10 m. (h) Distance: 2 m.

processed, the processing speed and the response speed of the proposed algorithm was increased.

To evaluate the performance of the proposed algorithm under real conditions, a camera was installed on a tractor (MF399) (Fig. 1). Performance was tested at 30 km/h under morning, noon and evening lighting conditions and five distances between the human and tractor using four replications. The proposed algorithm was combined with the HOG method [5] and the cascade method [13], as two by two and also all three methods together. The results were compared two-by-two and for a combination of the three methods for mode of composition and performance of fusion algorithms. The number of treatments was 15 and the total number of tests was 240. The results of the factorial completely randomized design were analyzed in SASv. 9.1 software. The results were compared for three indices:

Accuracy: The number of frames that were correctly detected as human by the number of frames that were correctly detected as non-human divided by total number of frames.

Pedestrian detection rate (PDR): The number of frames correctly detected as human divided by the number of frames correctly detected plus the number of frames incorrectly detected.

False alarm rate (FAR): The number of non-human frames incorrectly detected as human, divided by the number of frames that were correctly detected plus the number of frames in which a non-human was incorrectly detected as human.

The indices were based on the total number of frames available in the video and were calculated using Eqs. (5)–(7) as [7]:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \times 100 \quad (5)$$

$$\text{PDR} = \frac{TP}{TP + FN} \times 100 \quad (6)$$

$$\text{FAR} = \frac{FP}{FP + TP} \times 100 \quad (7)$$

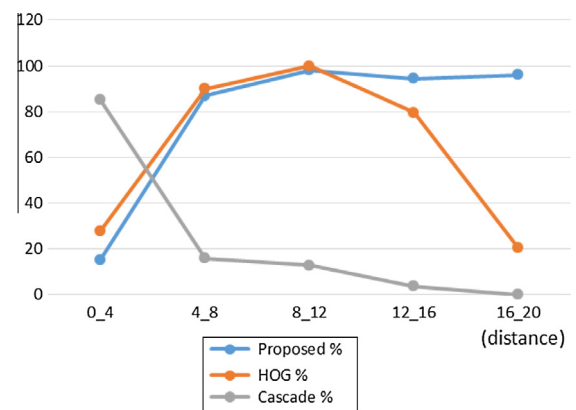


Fig. 10. Accuracy of proposed algorithm.

Table 1
Performance of proposed algorithm versus distance from human to tractor (%).

Evening			Noon			Morning			Distance (m)
FAR	PDR	Accuracy	FAR	PDR	Accuracy	FAR	PDR	Accuracy	
45	10	10	0	15	15	0	10	15	0–4
40	60	57	0	87	87	0	60	85	4–8
38	77	67	2	100	98.1	2	77	93	8–12
35	65	56	2.2	96.2	94.4	2.2	95	91.2	12–16
26	75	55	2.4	98.1	96	2.4	77	94	16–20

where TP is the number of detections in which humans were correctly identified, TN is the number of detections in which non-humans were correctly identified, FP is the number of detections

in which non-humans were incorrectly identified as human, and FN is the number of detections in which humans were incorrectly identified as non-humans.

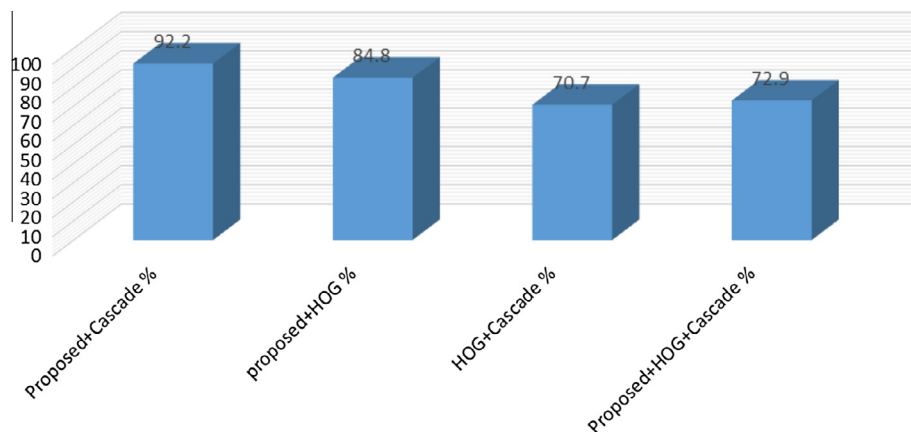


Fig. 11. Accuracy of combined algorithms.

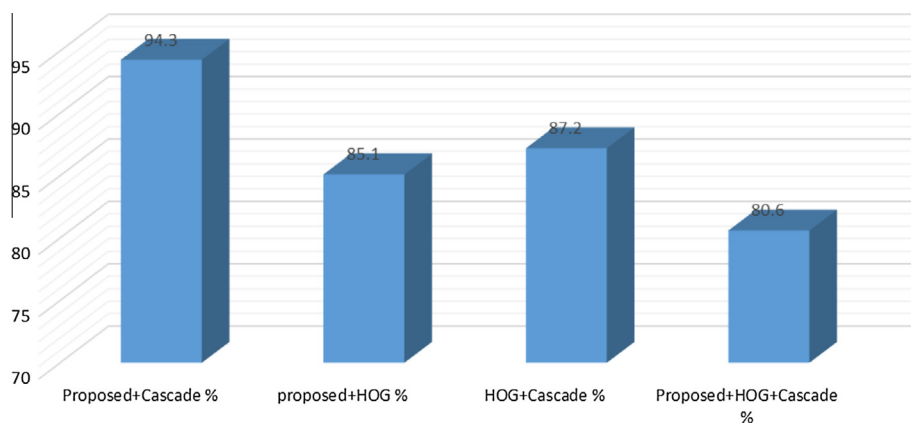


Fig. 12. PDR of combined algorithms.

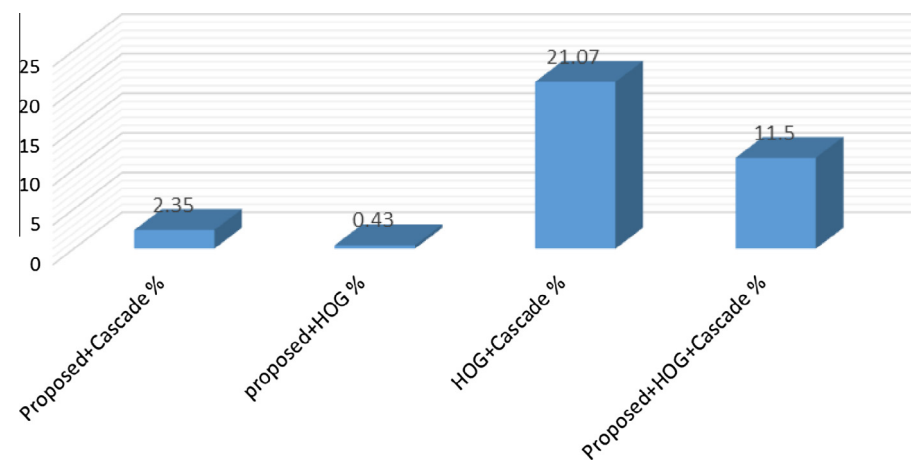


Fig. 13. FAR of combined algorithms.

Table 2
ANOVA for accuracy of different methods in human detection.

Source	df	Mean square
Method	3	3958.019**
Time	2	7.343 ns
Distance	4	5599.091**
Method * time	6	.228 ns
Method * distance	12	5250.459**
Time * distance	8	.245 ns
Method * time * distance	24	.164 ns
Error	180	4.459
Corrected total	239	

ns: non-significant.

** Significant at 1% level.

3. Results and discussion

Table 1 shows the results for the performance of the proposed at different distances from human to tractor. It can be seen that the performance in morning and evening lighting at 12–20 m provided the highest resolution compared to the other methods. The PDR and FAR at these distance was best; performance was also good at distances of 4–12 m.

The accuracy of the proposed algorithm was compared with those of the other algorithms at different distances. The cascade method showed the best accuracy at 0–4 m and showed good results at 4–12 m. At 8–20 m, the proposed algorithm showed good results (Fig. 10).

The proposed method was not accurate in comparison with the HOG and cascade methods at 0–4 m when a human is not in the frame in poor light conditions (shade). To produce an overall algorithm that is highly accurate at all distances under all lighting conditions, combinations using two-by-two and all three algorithms were carried out. Figs. 11–13 show the results of this procedure. Fig. 11 shows that the combination of the proposed algorithm with the cascade algorithm produced the highest accuracy at 92.2%. Fig. 12 shows that integrating the proposed method with the cascade method provided the best rate of detection of humans at 94.3%. Fig. 13 shows that combining the proposed method with the cascade method produced the lowest FAR at 2.35%.

The results of analysis of variance for accuracy of human identification are shown in Table 2. For the type of method versus

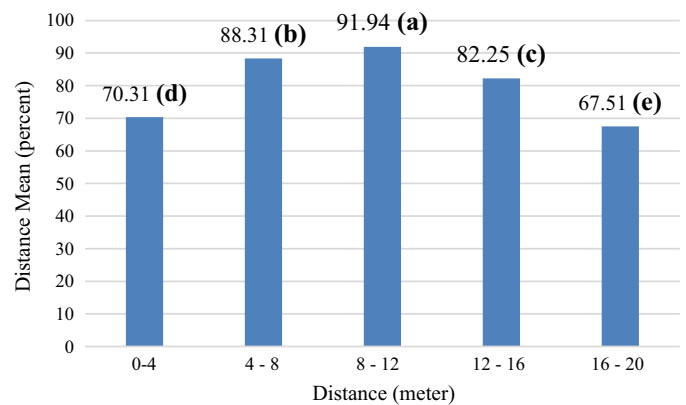


Fig. 15. Mean comparison of methods accuracy at different distances of Duncan test (same letters mean no significant difference).

distance from camera, accuracy was significant at 1% level, indicating that the performance of the methods differed from one another and the performance of each method differed according to distance. The dual interaction effect of identification method and distance from the camera was significant at 1% level; thus, the methods performed differently according to distance from human to tractor. Overall, an increase in distance and evening lighting conditions decreased the accuracy of the results.

The Results of mean comparison of vary compilation algorithms accuracy and also at different distances movement in Duncan test showed that the performance accuracy has significant difference and we will see different results at different distances from the performance (Figs. 14 and 15).

4. Conclusion

The results of this study indicated that combination of the proposed algorithm and Cascade method showed acceptable performance in identifying a human moving in front of a tractor. With compilation these algorithms, we will reach an optimal algorithm with acceptable performance at all distances and different lighting conditions. Better results for pedestrian detecting can be obtained by strengthening the algorithm in automatic guidance system for agricultural tractors.

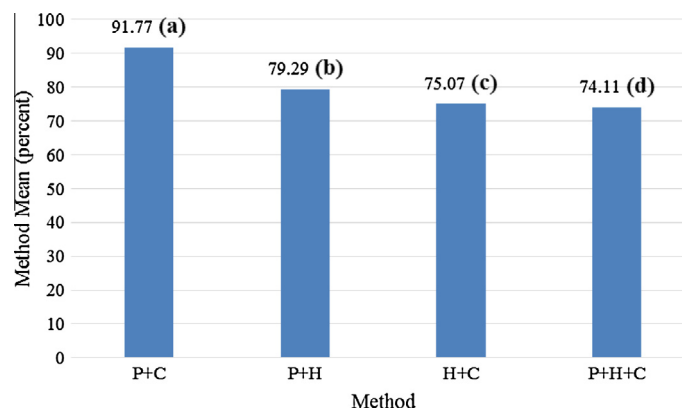


Fig. 14. Mean comparison of accuracy of the vary combined methods of Duncan test (same letters mean no significant difference).

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